Beyond Equilibrium: Quantifying and Explaining Inequality in LLM-Driven Data Markets with Privacy Choice Externalities

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Appendix A
Additional results for RQ2

Appendix B
Prompt for LLM-driven Multi-agent Data Market System

Table VIII presents the system prompts we designed for consumer and firm agents, while Table IV shows the prompts for firms when setting prices. The uppercase NOTE emphasizes guidance for firm agents regarding consumers' data sharing behavior, product purchasing decisions, and corporate revenue, aiding them in making more informed decisions.Tables VII, V , and III present the prompts for consumers regarding whether to continue searching, whether to share data, and whether to make a purchase, respectively.

TABLE III
Prompt for consumer purchasing decision

purchase_decision prompt
`{self.get_memory_for_prompt()}` You have the following valuations and prices for all firms: `{opts_str}`. The format is (Idx: Firm Index, Val: Your Valuation, Prc: Price Offered, Prft: Your Profit). Your historical decisions are: `{self.memory}`. Please choose a firm to purchase from or decide not to purchase. If you want to purchase, please provide the index of the firm. If you want to leave (stop searching and not purchase), please provide -1. Please provide your choice (only a number) and extremely brief reason. Format 1 (purchase)(Example which you must strictly follow its format): Decision: I'll purchase Firm 1 (choose number 1) because: - ... - ... Format 2 (leave) (Example which you must strictly follow its format): Decision: I'll leave (choose number -1) because: - ... - ...

TABLE IV
Pricing prompts used for firm

Pricing Prompts
Your goal is to maximize your total profit. Your cost is: `{self.firm_cost}` per product. The consumers' valuations for every firm's product are independently uniformly distributed between 0 and 1. There are `{self.num_firms}` firms in the market, including you. The more firms in the market, the more competition you will face. Your predicted share rate for this round is: `{self.share_rate_predicted}`. NOTE:THE MORE CONSUMERS SHARE DATA, THE MORE LIKELY THEY ARE TO RECEIVE PERSONALIZED RECOMMENDATIONS (EVERY FIRM HAS EQUAL CHANCE TO BE RECOMMENDED). THE MORE CONSUMERS RECEIVE PERSONALIZED RECOMMENDATIONS, THE LESS LIKELY THEY ARE TO SEARCH FOR OTHER FIRMS (IF OTHER FIRMS EXIST). THE LESS LIKELY THEY ARE TO SEARCH FOR OTHER FIRMS, THE MORE LIKELY THEY ARE TO BUY YOUR PRODUCT. THE MORE LIKELY THEY ARE TO BUY YOUR PRODUCT, THE MORE LIKELY YOU ARE TO SET A HIGHER PRICE. The more competition you face, the more valuable personalized recommendations become, as they help consumers find the best product more efficiently. With more firms, consumers are more likely to rely on personalized recommendations, reducing their search efforts and increasing their likelihood of purchasing your product. Your historical decisions are `{self.get_memory_for_prompt()}` Your basic price is: `{self.basic_price}`. Please provide a new price for your product and extremely brief reason. Format: 'Change price: xxx, reason: xxx'. Note: If you want to keep the basic price, then set the *Change price* to 0. If you want to increase the price, then set the *Change price* to a positive number. If you want to decrease the price, then set the *Change price* to a negative number. Be sure to keep the price positive and round to two decimal places (like 0.00).

TABLE V
Prompt for consumer sharing decision

share_decision prompt
`{self.get_memory_for_prompt()}` If you share your data, you will receive personalized recommendations. The personalized recommendations will be sorted by your valuation from high to low. The more firms in the market, the more valuable the recommendations will be. Now there are `{self.num_firms}` firms to choose from, which is the maximum number of firms you can search. Do you want to share your data? Please answer 'yes' or 'no' with your extremely brief reason.

TABLE I
AGGREGATE CONSUMER SURPLUS (↑) BY FIRM COUNT (n) UNDER HETEROGENEOUS DEPLOYMENTS

Setting		n=1	n=2	n=3	n=4	n=5	n=6	n=7	n=8	n=9	n=10
Homogeneous											
WW	weak firm + weak consumer	1.12	0.00	1.84	2.43	2.89	2.31	2.22	2.81	2.26	2.38
SS	strong firm + strong consumer	3.98	3.05	2.89	3.00	1.67	2.41	2.97	2.64	2.61	3.05
RR(BNE)	rational firm + rational consumer	2.50	*4.67*	*5.49*	3.73	2.28	2.18	2.07	1.97	1.87	1.78
Heterogeneous											
RW	rational firm + weak consumer	2.56	1.83	1.80	2.51	1.61	2.60	1.29	1.72	2.28	2.00
RS	rational firm + strong consumer	*3.48*	2.95	3.16	2.87	2.49	2.49	4.69	1.58	1.65	1.89
SW	strong firm + weak consumer	2.50	2.15	1.57	3.20	2.06	*4.70*	**6.92**	2.62	1.97	*5.79*
SR	strong firm + rational consumer	2.05	3.80	**6.11**	*3.79*	**6.21**	**6.56**	*6.51*	**6.38**	**7.23**	**7.30**
WS	weak firm + strong consumer	2.00	1.71	1.45	2.30	1.70	3.53	4.46	*4.46*	*2.24*	3.19
WR	weak firm + rational consumer	**3.54**	**5.84**	3.62	**3.95**	*3.60*	1.34	2.30	1.71	1.80	1.11

* Bold values denote the highest (best) consumer surplus, while underlined italicized values denote the second-best.

TABLE II
AGGREGATE SOCIAL WELFARE BY FIRM COUNT (n) UNDER HETEROGENEOUS DEPLOYMENTS

Setting		n=1	n=2	n=3	n=4	n=5	n=6	n=7	n=8	n=9	n=10
Homogeneous											
WW	weak firm + weak consumer	4.72	7.35	12.03	14.40	14.02	13.58	16.42	15.99	16.15	15.29
SS	strong firm + strong consumer	**11.47**	10.03	**13.51**	14.58	12.75	*16.40*	16.11	16.68	15.65	17.25
RR(BNE)	rational firm + rational consumer	7.50	8.24	8.23	6.32	4.61	4.24	3.93	3.66	3.42	3.21
Heterogeneous											
RW	rational firm + weak consumer	7.06	*11.07*	10.61	13.21	12.09	15.62	12.43	15.40	**17.77**	*17.74*
RS	rational firm + strong consumer	9.48	8.11	11.64	13.47	14.41	14.86	14.29	14.59	15.20	14.46
SW	strong firm + weak consumer	7.50	7.14	8.54	12.70	11.63	14.30	*17.02*	*16.70*	15.94	17.35
SR	strong firm + rational consumer	7.55	10.93	12.64	*15.24*	**16.21**	**16.56**	16.01	16.38	*17.23*	17.29
WS	weak firm + strong consumer	7.45	10.96	9.34	12.97	*15.17*	15.77	**17.19**	**17.97**	15.61	**17.96**
WR	weak firm + rational consumer	*10.54*	**11.84**	*13.49*	**15.40**	14.38	13.28	13.57	15.29	14.74	13.16

* Bold values denote the highest (best) social welfare, while underlined italicized values denote the second-best.

TABLE VI
PROMPTS USED FOR FIRM AND CONSUMER AGENTS

System Prompts
firm prompt: You are a firm. You will set the price of your product. Your goal is to maximize your profit. Your profit is the product of the number of consumers who choose to buy your product and the price, minus the cost. Please consider the number of consumers, the share rate of consumers who share data, and your historical decisions.
consumer prompt: You are a consumer. You will search for products to purchase. You have a privacy value of `{self.privacy_cost}`. There are `{self.num_firms}` firms to choose from. Please consider whether to share your data. If you share your data, you will receive personalized recommendations (your search sequence of products will be sorted from highest value to lowest value), which will reduce your total search cost. (The total search cost is the product of the number of searches and the search cost. Sharing your data can significantly reduce the number of searches to make a good deal.) If you don't share your data, you won't receive any recommendations, which will increase your total search cost on average. (In the worse case, the total search cost is the product of the number of searches and the search cost, which is `{self.num_firms - 1}` × `{self.search_cost}`.) Your goal is to maximize your total revenue. Your total revenue is the sum of the valuation of the product you purchase, minus the price of the product, minus the total search cost, minus the privacy value. When deciding whether to share your data, consider the trade-off between the privacy value and the potential reduction in search cost. If you choose to search more than a certain number, then it's impossible for you to get a positive profit. When deciding whether to search, consider the trade-off between the search cost and the potential revenue. Given the search cost is `{self.search_cost}` per search while the net profit from purchasing a product is less than 1, you should be cautious. The first search is free, but each subsequent search will cost you `{self.search_cost}`.

TABLE VII
PROMPT FOR CONSUMER SEARCHING DECISION

search_decision prompt

`{self.share and 'you have decided to share your data' or 'you have decided not to share your data'}`
You have searched `{len(self.searched_firms)}` firms.
The indices, valuations and prices of these firms are: `{searched_firms_list}`.
Format: (index, valuation, price)
And their potential profit (valuation - price) is: `{profits}`.
`{self.get_memory_for_prompt()}`
So far, your total search cost is `{self.total_search_cost}`.
You are rational: if any searched firm offers positive profit, you won't leave without buying.
Please choose one action:
1. Purchase: Choose the index of a firm you already searched (from the list above).
2. Search: Choose `{self.num_firms}` to search the next firm (this will cost `{platform.search_cost}`).
3. Leave: Choose -1 (stop searching, buy nothing).
Provide your choice (only the number) and extremely brief reason.
Format Examples (Strictly follow one):
Decision:
I'll purchase Firm X (choose number X) because:
- Reason 1
- Reason 2
Decision:
I'll search another firm (choose number `{self.num_firms}`) because:
- Reason 1
- Reason 2
Decision: I'll leave (choose number -1) because:
- Reason 1
- Reason 2

TABLE VIII
PROMPTS USED FOR FIRM AND CONSUMER AGENTS

System Prompts
firm prompt: You are a firm. You will set the price of your product. Your goal is to maximize your profit. Your profit is the product of the number of consumers who choose to buy your product and the price, minus the cost. Please consider the number of consumers, the share rate of consumers who share data, and your historical decisions.
consumer prompt: You are a consumer. You will search for products to purchase. You have a privacy value of `{self.privacy_cost}`. There are `{self.num_firms}` firms to choose from. Please consider whether to share your data. If you share your data, you will receive personalized recommendations (your search sequence of products will be sorted from highest value to lowest value), which will reduce your total search cost. (The total search cost is the product of the number of searches and the search cost. Sharing your data can significantly reduce the number of searches to make a good deal.) If you don't share your data, you won't receive any recommendations, which will increase your total search cost on average. (In the worse case, the total search cost is the product of the number of searches and the search cost, which is `{self.num_firms - 1}` × `{self.search_cost}`.) Your goal is to maximize your total revenue. Your total revenue is the sum of the valuation of the product you purchase, minus the price of the product, minus the total search cost, minus the privacy value. When deciding whether to share your data, consider the trade-off between the privacy value and the potential reduction in search cost. If you choose to search more than a certain number, then it's impossible for you to get a positive profit. When deciding whether to search, consider the trade-off between the search cost and the potential revenue. Given the search cost is `{self.search_cost}` per search while the net profit from purchasing a product is less than 1, you should be cautious. The first search is free, but each subsequent search will cost you `{self.search_cost}`.

APPENDIX C
ALGORITHM FOR SIMULATION FRAMEWORK

Algorithm 1 Heterogeneous Multi-Agent Simulation for Data Markets

Require: Environment $\mathcal{E} = \{N, n, s, c, T_{\max}\}$;
Valuation $v \sim F$ on $[\underline{v}, \overline{v}]$;
Privacy cost $\tau_i \sim T(\cdot)$;
Reservation r s.t. $\int_r^{\overline{v}} (1 - F(v))dv = s$;
Backends $\{\mathcal{B}_i^C\}$, $\{\mathcal{B}_j^F\}$;
Switches $\rho^{\text{Share}}, \rho^{\text{Price}}, \rho^{\text{Search}} \in \{0, 1\}$;
Functions: Net Benefit $\Delta(r, n)$, Demand $D_s(p_i, p), D_a(p_i, p)$.
Ensure: Series $\{\sigma^t, p^t_{\text{avg}}, SC^t, S^t_c, S^t_f, SW^t, R^t_{CS}\}_{t=1}^{T_{\max}}$.

Init: $\sigma^0 \sim U[0, 1]$, p^0_{avg}, $t \leftarrow 1$.
if Benchmark **then**
  Update $\sigma^0 \leftarrow \frac{1}{N} \sum \mathbb{I}\{\Delta(p^0_{\text{avg}}) > \tau_i\}$ iteratively until convergence.
end if
while $t \leq T_{\max}$ **do**
  // Stage 1: Consumer Sharing
  $I^t_{i,\text{share}} \leftarrow \mathbb{I}\{\Delta(r, n) > \tau_i\}$ **if** ρ^{Share} **else** $\texttt{LLM_Share}_i(\tau_i, n, s, r)$, $\forall i$.
  $\sigma^t \leftarrow \frac{1}{N} \sum_{i=1}^N I^t_{i,\text{share}}$
  // Stage 2: Firm Pricing
  if $\rho^{\text{Price}} = 1$ **then**
    Initialize guess prices $\{p_j\}_{j=1}^n$
    repeat
      $p_{\text{avg}} \leftarrow \frac{1}{n} \sum p_j$
      for $j = 1$ **to** n **do**
        $p^{\text{new}}_j \leftarrow \arg\max_{p \geq c} (p - c)[\sigma^t D_s(p, p_{\text{avg}}) + (1 - \sigma^t) D_a(p, p_{\text{avg}})]$
      end for
      Update prices $\{p_j\} \leftarrow \{p^{\text{new}}_j\}$
    until prices converge
    $p^t_j \leftarrow p_j, \forall j$
  else
    for $j = 1$ **to** n **do**
      $p^t_j \leftarrow \texttt{LLM_Price}_j(\sigma^t, n, c)$
    end for
  end if
  $p^t_{\text{avg}} \leftarrow \frac{1}{n} \sum_{j=1}^n p^t_j$
  // Stage 3: Search and Purchase
  for $i = 1$ **to** N **do**
    Gen $\mathbf{v}_i$; $\mathcal{O}_i \leftarrow \text{sort}(\{j\}, v_{ij})$ if $I^t_{i,\text{share}}$ else $\text{permute}(\{j\})$.
    if $\rho^{\text{Search}} = 1$ **then**
      Search $\mathcal{O}_i$ for first j^* s.t. $v_{ij} - p^t_j \geq r - p^t_{\text{avg}}$.
      if j^* exists **then** Buy j^*; **else** Buy $\arg\max(v - p)$ if > 0, else Leave.
    else
      $\text{Action}_i \leftarrow \texttt{LLM_SearchBuy}_i(\mathcal{O}_i, \{p^t_j\}_{j \in \mathcal{O}_i}, r, s)$.
    end if
  end for
  Compute Metrics $\{\sigma^t, p^t_{\text{avg}}, SC^t, S^t_c, S^t_f, SW^t, R^t_{CS}\}$; $t \leftarrow t + 1$.
end while

APPENDIX D
PROMPTS FOR CORE FRAMEWORK

Table IX and Table XI represent the prompts for the Decision Factor Importance Ranking (Agent_{rank}), while Table X and Table XII represent the prompts for Decision Chain (Agent_{matrix}). Table XIII and Table XIV represent the prompts for the expert agent (Agent_{audit}) when judging the validity of the extraction. Table XV presents the prompts for the extraction and supplementation of decision factors ($\text{Agent}_{extract}$).

TABLE IX
$Agent_{rank}$ CONSUMER FACTOR PROMPT CONFIGURATION

Title: $Agent_{rank}$ consumer factor prompt

Content:

You are a structured data analysis assistant. Please read the following single line of CSV data and summarize two outputs according to our factor rules below:
Step 1: For each Factor, list the evidence found in the file.
Step 2: Based on the evidence, evaluate and assign an importance ranking (ties allowed, three levels in total: 1 = directly drives decision, 2 = secondary influence, 3 = marginal impact). The generated ranking must include 1, 2, and 3, with roughly equal proportions.
Step 3: Output a pure vector (no extra text), with a fixed dimension equal to the number of factors (8D), formatted as: [1,2,1,3,2,1,3,2]. The order of factors follows the table below.
Rows correspond to factor pairs, with both rows and columns ordered according to the table below.

Major Category	Factor	Definition	Influence on Sharing Decision	Influence on Purchase Decision
Benefits	Search Cost Savings	Reduction in search cost due to personalized recommendations after sharing	High savings promote sharing ("significant reduction...outweighs privacy" → `share=True`); low savings suppress sharing ("minimal...privacy outweighs" → `share=False`).	High savings promote `BestProfit` and encourage purchase ("positive 0.21...purchase"); low savings push search/exit ("minimal...negative...leave").
	Positive Profit	Perceived positive profit	Positive profit promotes sharing ("expect positive...outweighs privacy" → `share=True`); lack of positive profit suppresses sharing ("no positive...privacy outweighs" → `share=False`).	Positive profit promotes purchase ("positive 0.23...buy"); lack of profit drives search/exit ("no positive...leave").
Drawbacks	Privacy Loss	Value and immediate perception of privacy loss	High loss suppresses sharing ("privacy 0.051 incurs...outweighs" → `share=False`); low loss promotes sharing ("low privacy 0.026...share" → `share=True`).	High loss + negative profit drive exit ("privacy 0.055...negative...leave"); low loss promotes purchase ("low privacy...positive...buy").
	Negative Profit	Perceived negative profit	Negative profit suppresses sharing ("expect negative...privacy outweighs" → `share=False`); no negative profit promotes sharing ("no negative...share" → `share=True`).	Negative profit drives exit ("negative...leave"); no negative profit promotes purchase ("positive...buy"); uncertainty drives search ("zero...search further").
	Search Continuation Cost	Additional cost of continued search (e.g., "incurs 0.02 per search...no improvement")	High cost promotes sharing ("share to avoid 0.02 costs" → `share=True`); low cost reduces sharing incentive.	High cost promotes purchase/exit ("incurs 0.02...purchase positive"); low cost promotes search ("low cost...search further").
Trade-offs	Net Benefit	Compare savings vs. privacy cost to calculate `BestProfit` net value (e.g., "outweighs" + "positive/negative profit")	Positive net promotes sharing ("significant outweighs privacy...positive net" → `share=True`); negative net suppresses sharing ("privacy outweighs...negative net" → `share=False`).	Positive net benefit promotes purchase ("positive 0.23...purchase"); negative net benefit drives exit/search ("negative -0.14...leave").
	Profit Maximization	Integration of pros and cons based on "maximize revenue"	Rational behavior promotes sharing ("rational to share...maximize" → `share=True`); suppresses sharing ("not share to maximize").	Rational behavior promotes purchase ("rational...buy positive"); drives exit ("rational...leave").
	Loss Minimization	Integration of pros and cons based on "minimize loss"	High loss suppresses sharing ("rational to minimize...not share" → `share=False`); low loss promotes sharing ("no loss...share").	Rational behavior drives exit/search ("minimize loss...leave"; "zero...search further").

{ "result": "<your extracted importance vector>" }

TABLE X
$Agent_{matrix}$ CONSUMER LOGIC PROMPT CONFIGURATION

Title: $Agent_{matrix}$ consumer Logic prompt

Content:

You are a structured data analysis assistant. Please read the following single line of CSV data and summarize two outputs according to our factor rules below:
Step 1: For each Factor, list the evidence found in the file.
Step 2: Based on the above factor analysis, first list the relationships supported by evidence, then generate an 8×8 undirected adjacency matrix (the matrix must be symmetric), representing the logical relationships between factors.
Values: 0 = no significant logical relationship, 1 = clear logical relationship
Format: [[0,1,0,...],[1,0,...],...]
Rows correspond to factor pairs, with both rows and columns ordered according to the table below.

Major Category	Factor	Definition	Influence on Sharing Decision	Influence on Purchase Decision
Benefits	Search Cost Savings	Reduction in search cost due to personalized recommendations after sharing	High savings promote sharing ("significant reduction...outweighs privacy" → `share=True`); low savings suppress sharing ("minimal...privacy outweighs" → `share=False`).	High savings promote `BestProfit` and encourage purchase ("positive 0.21...purchase"); low savings push search/exit ("minimal...negative...leave").
	Positive Profit	Perceived positive profit	Positive profit promotes sharing ("expect positive...outweighs privacy" → `share=True`); lack of positive profit suppresses sharing ("no positive...privacy outweighs" → `share=False`).	Positive profit promotes purchase ("positive 0.23...buy"); lack of profit drives search/exit ("no positive...leave").
Drawbacks	Privacy Loss	Value and immediate perception of privacy loss	High loss suppresses sharing ("privacy 0.051 incurs...outweighs" → `share=False`); low loss promotes sharing ("low privacy 0.026...share" → `share=True`).	High loss + negative profit drive exit ("privacy 0.055...negative...leave"); low loss promotes purchase ("low privacy...positive...buy").
	Negative Profit	Perceived negative profit	Negative profit suppresses sharing ("expect negative...privacy outweighs" → `share=False`); no negative profit promotes sharing ("no negative...share" → `share=True`).	Negative profit drives exit ("negative...leave"); no negative profit promotes purchase ("positive...buy"); uncertainty drives search ("zero...search further").
	Search Continuation Cost	Additional cost of continued search (e.g., "incurs 0.02 per search...no improvement")	High cost promotes sharing ("share to avoid 0.02 costs" → `share=True`); low cost reduces sharing incentive.	High cost promotes purchase/exit ("incurs 0.02...purchase positive"); low cost promotes search ("low cost...search further").
Trade-offs	Net Benefit	Compare savings vs. privacy cost to calculate `BestProfit` net value (e.g., "outweighs" + "positive/negative profit")	Positive net promotes sharing ("significant outweighs privacy...positive net" → `share=True`); negative net suppresses sharing ("privacy outweighs...negative net" → `share=False`).	Positive net benefit promotes purchase ("positive 0.23...purchase"); negative net benefit drives exit/search ("negative -0.14...leave").
	Profit Maximization	Integration of pros and cons based on "maximize revenue"	Rational behavior promotes sharing ("rational to share...maximize" → `share=True`); suppresses sharing ("not share to maximize").	Rational behavior promotes purchase ("rational...buy positive"); drives exit ("rational...leave").
	Loss Minimization	Integration of pros and cons based on "minimize loss"	High loss suppresses sharing ("rational to minimize...not share" → `share=False`); low loss promotes sharing ("no loss...share").	Rational behavior drives exit/search ("minimize loss...leave"; "zero...search further").

{"Matrix": "<your extracted full adjacency matrix>" }

TABLE XI
$Agent_{rank}$ FIRM FACTOR PROMPT CONFIGURATION

<table>
<tr><td>Title</td><td>$Agent_{rank}$ firm Factor prompt</td></tr>
<tr><td>Content</td><td>You are a structured data analysis assistant. Please read the following single line of CSV data and summarize two outputs according to our factor rules below:
Step 1: For each Factor, list the evidence found in the file.
Step 2: Based on the evidence, evaluate and assign an importance ranking (ties allowed, three levels in total: 1 = directly drives decision, 2 = secondary influence, 3 = marginal impact). The generated ranking must include 1, 2, and 3, with roughly equal proportions.
Step 3: Output a pure vector (no extra text), with a fixed dimension equal to the number of factors (8D), formatted as: [1,2,1,3,2,1,3,2]. The order of factors follows the table below.
Rows correspond to factor pairs, with both rows and columns ordered according to the table below.
<table>
<tr><td>Major Category</td><td>Factor</td><td>Definition</td><td>Influence on Pricing Decision (High/Mid/Low Pricing)</td></tr>
<tr><td>Benefits</td><td>Profit Margin</td><td>Increase in unit profit from higher pricing</td><td>High margin drives high price (“high margin...maximizes profit” → high); low margin drives low price (“low margin...reduce price to attract” → low).</td></tr>
<tr><td></td><td>Customer Loyalty</td><td>Dependence of data-sharing consumers on recommendations</td><td>Loyalty drives high price (“loyal sharing base...sustain high price” → high); low loyalty drives low price (“no loyalty...lower price to compete” → low).</td></tr>
<tr><td>Drawbacks</td><td>Competitive Pressure</td><td>Market competition leading to price wars</td><td>High competition suppresses high prices (“intense competition...reduce price” → low); low competition allows high prices (“no competition...high price” → high).</td></tr>
<tr><td></td><td>Customer Churn Risk</td><td>Risk of customers leaving due to high pricing</td><td>High risk suppresses high price (“leave risk...lower price” → low); low risk allows high price (“low leave risk...high price” → high).</td></tr>
<tr><td></td><td>Demand Uncertainty</td><td>Demand fluctuation leading to pricing errors</td><td>High uncertainty suppresses high price (“uncertain...adjust to mid price” → mid); low uncertainty drives high price (“stable demand...high price” → high).</td></tr>
<tr><td>Trade-offs</td><td>Net Profit</td><td>Compare pricing returns vs. cost risk to calculate expected profit</td><td>Positive net drives high price (“high outweighs low risk...positive net” → high); negative net drives low price (“competition outweighs...negative net” → low).</td></tr>
<tr><td></td><td>Profit Maximization</td><td>Integration of pros and cons based on “rational...maximize profit”</td><td>Drives high price (“rational to high price...maximize” → high); suppresses high price (“not high to maximize” → low).</td></tr>
<tr><td></td><td>Loss Minimization</td><td>Integration of pros and cons based on “rational...minimize loss”</td><td>Suppresses high price (“rational to minimize...not high price” → low); drives high price (“no loss...high price”).</td></tr>
</table>
Please output only JSON in the following format:
{ “result”: “<your extracted importance vector>”}</td></tr>
</table>

TABLE XII
$Agent_{matrix}$ FIRM LOGIC PROMPT CONFIGURATION

Title $Agent_{matrix}$ firm Logic prompt

Content

You are a structured data analysis assistant. Please read the following single line of CSV data and summarize two outputs according to our factor rules below:
Step 1: For each Factor, list the evidence found in the file.
Step 2: Based on the above factor analysis, first list the relationships supported by evidence, then generate an 8×8 undirected adjacency matrix (the matrix must be symmetric), representing the logical relationships between factors.
Values: 0 = no significant logical relationship, 1 = clear logical relationship
Format: [[0,1,0,...],[1,0,...],...]
Rows correspond to factor pairs, with both rows and columns ordered according to the table below.

Major Category	Factor	Definition	Influence on Pricing Decision (High/Mid/Low Pricing)
Benefits	Profit Margin	Increase in unit profit from higher pricing	High margin drives high price ("high margin...maximizes profit" → high); low margin drives low price ("low margin...reduce price to attract" → low).
	Customer Loyalty	Dependence of data-sharing consumers on recommendations	Loyalty drives high price ("loyal sharing base...sustain high price" → high); low loyalty drives low price ("no loyalty...lower price to compete" → low).
Drawbacks	Competitive Pressure	Market competition leading to price wars	High competition suppresses high prices ("intense competition...reduce price" → low); low competition allows high prices ("no competition...high price" → high).
	Customer Churn Risk	Risk of customers leaving due to high pricing	High risk suppresses high price ("leave risk...lower price" → low); low risk allows high price ("low leave risk...high price" → high).
	Demand Uncertainty	Demand fluctuation leading to pricing errors	High uncertainty suppresses high price ("uncertain...adjust to mid price" → mid); low uncertainty drives high price ("stable demand...high price" → high).
Trade-offs	Net Profit	Compare pricing returns vs. cost risk to calculate expected profit	Positive net drives high price ("high outweighs low risk...positive net" → high); negative net drives low price ("competition outweighs...negative net" → low).
	Profit Maximization	Integration of pros and cons based on "rational...maximize profit"	Drives high price ("rational to high price...maximize" → high); suppresses high price ("not high to maximize" → low).
	Loss Minimization	Integration of pros and cons based on "rational...minimize loss"	Suppresses high price ("rational to minimize...not high price" → low); drives high price ("no loss...high price").

Please output only JSON in the following format:
{"Matrix": "<your extracted full adjacency matrix>" }

TABLE XIII
$Agent_{audit}$ CONSUMER PROMPT CONFIGURATION

Title $Agent_{audit}$ consumer prompt

Content

You are a rigorous annotation expert specializing in "decision logic", responsible for verifying whether the [Decision Factor Importance Vector] and [Factor Relationship Adjacency Matrix] provided by another model (Agent 1) are consistent with the textual rationale.

1. Explanation of Decision Factors (fixed order; all vectors/matrices follow this sequence):

- **F1:** Search Cost Savings — Search cost savings resulting from privacy sharing / search behavior
- **F2:** Positive Profit — Importance of positive profit (> 0)
- **F3:** Privacy Loss — Privacy loss or privacy risk
- **F4:** Negative Profit — Negative profit (< 0) or losses
- **F5:** Search Continuation Cost — Cost of continuing the search process
- **F6:** Net Benefit — Net benefit (benefits minus costs/risks)
- **F7:** Profit Maximization — Objective of profit maximization
- **F8:** Loss Minimization — Loss minimization / risk aversion

2. Input Text and Scenario Information

You will see key information about the consumer: Whether privacy was shared and the rationale (`share_reason`); Final decision to purchase/leave and the rationale (`final_reason`). Collectively referred to as **[Raw Input Data]**.

`[Raw Input Data]: {input_data}`

3. Candidate Annotation Results from Agent 1

1. *importance_vector*: List of length 8 (F1–F8). Values: 1=Primary, 2=Secondary, 3=Weak/Irrelevant.
2. *adjacency_matrix*: 8×8 binary matrix. `matrix[i][j]=1` indicates a clear decision relationship between F_i and F_j.

`[Extraction Results to be Reviewed]: {output_data}`

4. Your Review Task and Criteria

Determine if results are "generally reasonable" based **only** on [Raw Input Data].
- **Output 1 (Valid):** Align with rationale; minor discrepancies acceptable.
- **Output 0 (Invalid):** Do not align, logical fallacies, or significant errors.

5. Output Format

- Output ONLY a single number: 0 or 1.
- Do NOT output any explanations or additional comments.

TABLE XIV
$Agent_{audit}$ FIRM PROMPT CONFIGURATION

Title	$Agent_{audit}$ firm prompt

Content	You are a rigorous annotation expert specializing in "decision logic", responsible for verifying whether the [Decision Factor Importance Vector] and [Factor Relationship Adjacency Matrix] provided by another model (Agent 1) are consistent with the textual rationale.
	1. Explanation of Decision Factors (fixed order; all vectors/matrices follow this sequence):
	F1: Profit Margin : Increase in unit profit from higher pricing.
	F2: Customer Loyalty : Dependence of data-sharing consumers on recommendations.
	F3: Competitive Pressure : Market rivalry driving price competition.
	F4: Customer Churn Risk : Risk of customers leaving due to high pricing.
	F5: Demand Uncertainty : Demand fluctuation leading to pricing errors.
	F6: Net Profit : Comparison (pricing returns vs. cost risk).
	F7: Profit Maximization : Drive toward maximizing revenue.
	F8: Loss Minimization : Drive toward mitigating risk/loss.
	2. Input Text and Scenario Information
	You will be provided with key information about the company: Price adjustments and pricing rationale (`price_reason`). Collectively referred to as **[Raw Input Data]**.
	`[Raw Input Data]: {input_data}`
	3. Candidate Annotation Results from Agent 1
	1. *importance_vector*: List of length 8 (F1–F8). Values: 1=Primary, 2=Secondary, 3=Weak/Irrelevant.
	2. *adjacency_matrix*: 8×8 binary matrix. `matrix[i][j]=1` indicates a clear decision-making relationship between Factor F_i and Factor F_j.
	`[Extraction Results to be Reviewed]: {output_data}`
	4. Your Review Task and Criteria
	Determine if results are "generally reasonable" based **only** on the textual rationale in the [Raw Input Data].
	- **Output 1 (Valid):** Align with rationale; minor discrepancies acceptable.
	- **Output 0 (Invalid):** Do not align, logical fallacies, or significant errors.
	5. Output Format
	- Output ONLY a single number: 0 or 1.
	- Do NOT output any explanations or additional comments.

TABLE XV
PROMPT FOR INDUCTIVE CODING AND TAXONOMY CONSTRUCTION

Component	**Content**
Role	You are an expert researcher in Behavioral Economics and Data Privacy.
Task	Perform an Inductive Coding process to extract decision factors from LLM Agent rationales.
Context	We are building a taxonomy of factors influencing data market decisions. Our initial framework includes: `{initial_framework}`.
Input	`"{agent_rationale}"`
Instructions	1. Analyze the input rationale to identify decision drivers. 2. Validate if the initial factors exist in this rationale. 3. Inductively identify any emerging factors not covered by the initial list. 4. Categorize all identified factors into three dimensions: Benefits, Drawbacks, and Trade-offs.

Output Format	Please output in the following JSON structure:
	`{`
	`  "validated_factors": [],`
	`  "emerging_factors": [],`
	`  "final_taxonomy_segment": {`
	`  "Benefits": [],`
	`  "Drawbacks": [],`
	`  "Trade-offs": []`
	`  },`
	`  "evidence_quote": "..."`
	`}`